

EDUCATION

National University of Singapore

Ph.D. in Mathematics, Advisor: Tan Minh Nguyen

Singapore

2022–Present

Vietnam National University – University of Science

B.A. in Mathematics, with Honors

Hanoi, Vietnam

2018–2022

RESEARCH INTERESTS

- **Symmetry in Parameter Space.** I study a central question: given a model f , how are parameter settings θ and $\bar{\theta}$ related when they represent the same function, $f_\theta \equiv f_{\bar{\theta}}$? Applications include understanding implicit bias through neural conservation laws, symmetry-aware optimization and quantization, equivariant metanetworks, and mode connectivity up to symmetry.
- **AI for Mathematical Reasoning.** I am also interested in using AI to tackle mathematical problems, particularly in commutative algebra and algebraic geometry, to better understand the capabilities and limitations of large language models in mathematical reasoning.
I share updates on this work at <https://vh-tran.github.io/ai-in-maths.html>.

PUBLICATIONS

[†] denotes first or co-first authorship.

Conference Publications

1. **Conservation Laws for Modern Neural Architectures**
Viet-Hoang Tran[†], Vinh Khanh Bui[†], Tan Lai Ngoc, Nam Nguyen, Tuan Quang Dam, Tan Minh Nguyen.
International Conference on Machine Learning (ICML), 2026 (spotlight).
2. **Functional Equivalence in Attention: A Comprehensive Study with Applications to Linear Mode Connectivity**
Viet-Hoang Tran[†], Khanh Vinh Bui, Van-Hoan Trinh, Tan Lai Ngoc, Tan Minh Nguyen.
International Conference on Machine Learning (ICML), 2026.
3. **Geometric and Stochastic Analysis of Discontinuities in Sparse Mixture-of-Experts**
Tho Tran Huu[†], Huu-Tuan Nguyen, Thien-Hai Nguyen, Nhat-Tri Ho, **Viet-Hoang Tran**, Tho Quan, Tan Minh Nguyen.
International Conference on Machine Learning (ICML), 2026 (spotlight).
4. **Quasi-Equivariant Metanetworks**
Viet-Hoang Tran[†], An Nguyen The[†], Benoit Gruerand, Thieu Vo, Tan Minh Nguyen.
International Conference on Learning Representations (ICLR), 2026.
5. **Mixed-Curvature Tree-Sliced Wasserstein Distance**
Duy-Tung Pham[†], **Viet-Hoang Tran**[†], Thieu Vo, Tan Minh Nguyen.
International Conference on Learning Representations (ICLR), 2026.
6. **Tree-sliced Sobolev IPM**
Viet-Hoang Tran[†], Thanh Tran[†], Thanh Chu, Duy-Tung Pham, Trung-Khang Tran, Tam Le, Tan Minh Nguyen.
International Conference on Learning Representations (ICLR), 2026.
7. **Revisiting Tree-Sliced Wasserstein Distance Through the Lens of the Fermat-Weber Problem**
Viet-Hoang Tran[†], Thanh Tran[†], Thanh Chu[†], Duy-Tung Pham, Trung-Khang Tran, Tam Le, Tan Minh Nguyen.
International Conference on Learning Representations (ICLR), 2026.
8. **On Linear Mode Connectivity of Mixture-of-Experts Architectures**
Viet-Hoang Tran[†], Van Hoan Trinh[†], Khanh Vinh Bui[†], Tan M. Nguyen.
Advances in Neural Information Processing Systems (NeurIPS), 2025 (oral).

9. **Tree-Sliced Entropy Partial Transport**

Viet-Hoang Tran[†], Thanh Tran, Thanh Chu, Tam Le, Tan Minh Nguyen.

Advances in Neural Information Processing Systems (NeurIPS), 2025.

10. **Dynamical Properties of Tokens in Self-Attention and Effects of Positional Encoding**

Duy-Tung Pham[†], An Nguyen The[†], **Viet-Hoang Tran**, Nhan-Phu Chung, Xin T. Tong, Tan Minh Nguyen, Thieu Vo.

Advances in Neural Information Processing Systems (NeurIPS), 2025.

11. **Tree-Sliced Wasserstein Distance with Nonlinear Projection**

Thanh Tran[†], **Viet-Hoang Tran**[†], Thanh Chu, Trang Pham, Laurent El Ghaoui, Tam Le, Tan M. Nguyen.

International Conference on Machine Learning (ICML), 2025.

12. **Tree-Sliced Wasserstein Distance: A Geometric Perspective**

Viet-Hoang Tran[†], Trang Pham[†], Tho Tran, Minh Khoi Nguyen Nhat, Thanh Chu, Tam Le, Tan M. Nguyen.

International Conference on Machine Learning (ICML), 2025.

13. **Equivariant Polynomial Functional Networks**

Thieu N. Vo[†], **Viet-Hoang Tran**[†], Tho Tran Huu[†], An Nguyen The, Thanh Tran, Minh-Khoi Nguyen-Nhat, Duy-Tung Pham, Tan Minh Nguyen.

International Conference on Machine Learning (ICML), 2025.

14. **Equivariant Neural Functional Networks for Transformers**

Viet-Hoang Tran[†], Thieu N. Vo[†], An Nguyen The[†], Tho Tran Huu, Minh-Khoi Nguyen-Nhat, Thanh Tran, Duy-Tung Pham, Tan Minh Nguyen.

International Conference on Learning Representations (ICLR), 2025.

15. **Distance-Based Tree-Sliced Wasserstein Distance**

Viet-Hoang Tran[†], Khoi N.M. Nguyen[†], Trang Pham, Thanh T. Chu, Tam Le, Tan M. Nguyen.

International Conference on Learning Representations (ICLR), 2025.

16. **Spherical Tree-Sliced Wasserstein Distance**

Viet-Hoang Tran[†], Thanh T. Chu[†], Khoi N.M. Nguyen, Trang Pham, Tam Le, Tan M. Nguyen.

International Conference on Learning Representations (ICLR), 2025.

17. **Monomial Matrix Group Equivariant Neural Functional Networks**

Viet-Hoang Tran[†], Thieu N. Vo[†], Tho Tran-Huu, An T. Nguyen, Tan M. Nguyen.

Advances in Neural Information Processing Systems (NeurIPS), 2024.

Workshop Publications

1. **Modeling Expert Interactions in Sparse Mixture of Experts via Graph Structures**

Minh-Khoi Nguyen-Nhat[†], Rachel S.Y. Teo, Laziz Abdullaev, Maurice Mok, **Viet-Hoang Tran**, Tan Minh Nguyen.

AAAI 2026 Workshop MATH4AI.

2. **Equivariant Neural Functional Networks for Transformers**

Viet-Hoang Tran[†], Thieu N. Vo[†], An Nguyen The[†], Tho Tran Huu, Minh-Khoi Nguyen-Nhat, Thanh Tran, Duy-Tung Pham, Tan Minh Nguyen.

ICLR 2025 Workshop Weight Space Learning (spotlight).

3. **Distance-Based Tree-Sliced Wasserstein Distance**

Viet-Hoang Tran[†], Khoi N.M. Nguyen[†], Trang Pham, Thanh T. Chu, Tam Le, Tan M. Nguyen.

ICLR 2025 DeLTa Workshop.

4. **Spherical Tree-Sliced Wasserstein Distance**

Viet-Hoang Tran[†], Thanh T. Chu[†], Khoi N.M. Nguyen, Trang Pham, Tam Le, Tan M. Nguyen.

ICLR 2025 DeLTa Workshop.

Preprints

1. **A Clifford algebraic approach to $E(n)$ -equivariant high-order graph neural networks**
Viet-Hoang Tran[†], Thieu N. Vo[†], Tho Tran, Tan Nguyen.
preprint.

HONORS AND AWARDS

• Graduate Tutor Merit Award (Tier 1)	2026
• ICML 2026 Gold Reviewer	2026
• Graduate Tutor Merit Award (Tier 1)	2025
• NUSGS Research Incentive Award	2023
• NUS President's Graduate Fellowship	2022
• International Mathematics Competition for University Students, <i>Grand First Prize</i>	2020
• Vietnam Mathematical Olympiad for University Students, <i>First Prizes in Algebra and Analysis</i>	2019
• International Mathematical Olympiad, <i>Bronze Medal</i>	2018
• Vietnam Mathematical Olympiad for High School Students, <i>First Prize</i>	2018

TEACHING EXPERIENCE

National University of Singapore

MA2104 Multivariable Calculus	Spring 2026
MA2104 Multivariable Calculus	Fall 2025
DSA2102 Essential Data Analytics Tools: Numerical Computation	Spring 2025
MA2311 Techniques in Advanced Calculus	Fall 2024

Tran Phu High School for Gifted Students

Mathematics for Vietnam Mathematical Olympiad	Fall 2018–Spring 2023
Mathematics for Vietnam IMO Team Selection Test	Fall 2018–Fall 2022

TALKS AND ACADEMIC ACTIVITIES

• Talk: <i>Symmetry in Parameter Space</i> Workshop on Mathematical Approaches to Modern AI Challenges, Institute for Mathematical Sciences, National University of Singapore	Sep 2026
• Talk: <i>Symmetry in Parameter Space</i> Workshop on Mathematical Foundations of AI Models, Institute for Mathematical Sciences, National University of Singapore	Aug 2026
• Poster presentation International Conference on Machine Learning 2026, Seoul, South Korea	Jul 2026
• Talk: <i>Functional Equivalence in Neural Networks</i> Weekly Seminar, Department of Mathematics, University of California, Los Angeles	Apr 2026
• Poster presentation International Conference on Learning Representations 2026, Rio de Janeiro	Apr 2026
• Talk: <i>Functional Equivalence in Neural Networks</i> Hanoi–Singapore Meeting, Vietnam Institute for Advanced Study in Mathematics, Hanoi	Apr 2026
• Talk: <i>Symmetry in Parameter Space</i> Graduation Day, Information Theory and Applications Workshop 2026, San Diego	Feb 2026

- Talk: *Symmetry in Parameter Space: Theory and Application* Dec 2025
Machine Learning Seminar, SOICT-HUST, Hanoi University of Science and Technology
- Oral presentation: *On Linear Mode Connectivity of Mixture-of-Experts Architectures* Dec 2025
Poster presentation
Conference on Neural Information Processing Systems 2025, San Diego
- Talk: *Functional Equivalence in Neural Architectures* Jul 2025
Mathematical Seminar, Faculty of Mathematics, Hanoi University of Science
- Poster presentation Jul 2025
International Conference on Machine Learning 2025, Vancouver
- Talk: *Functional Equivalence in Neural Architectures* Jun 2025
Joint FDU-NUS-ZJU PhD Forum in Mathematics, National University of Singapore
- Talk: *Equivariant Neural Functional Networks for Transformers* Apr 2025
ICLR 2025 Workshop on Weight Space Learning, Singapore
- Poster presentation Apr 2025
International Conference on Learning Representations 2025, Singapore
- Information Theory and Applications Workshop 2025, San Diego Feb 2025
- Poster presentation Dec 2024
Conference on Neural Information Processing Systems 2024, Vancouver
- Talk: *Algebraic Perspective of Neural Architecture* Oct 2024
Seminar, Department of Electrical and Computer Engineering, Rice University
- SIAM Conference on Mathematics of Data Science 2024, Atlanta Oct 2024
- International Conference on Machine Learning 2024, Vienna Jul 2024
- Poster presentation Mar 2024
Workshop on Optimal Transport, Berlin
- The 62nd IEEE Conference on Decision and Control, Singapore Dec 2023
- Seminars in Algebraic Geometry, Vietnam Institute of Mathematics 2019–2022
- Talk: *Introduction to Elliptic Curves* Aug 2020
Workshop on The Mordell-Weil Theorem, Hanoi
- Boundary Behaviour of Holomorphic Mapping May–Jul 2020
Scientific Research for University Students Program, Vietnam Institute of Mathematics
- SEAMS-School on Arithmetic, Geometry and Model Theory, Hanoi Feb 2020
- IMH-School on Introduction to Algebraic Schemes and Cohomology, Hanoi Nov 2019